

MAT 534 HW09

Aditya Pahuja

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Problem 1. A module M is said to be **simple** if M is not the zero module and the only submodules of M are M and 0 . Show that M is a simple module if and only if $M \cong R/I$, where I is a maximal ideal of R (where isomorphism is as an R -module).

Solution. Let m be any nonzero element of simple module M . Then $\{rm : r \in R\}$ is a nontrivial submodule of M , so it equals M . Let I be the annihilator of m , which by Problem 4(a) is an ideal of R ; then, by (my solution to) Problem 4(b), $M \cong R/I$ with isomorphism $\phi_m : R/I \rightarrow M$ given by $\phi_m(r+I) = rm$. This means that given nonzero $s+I \in R/I$, there exists $r+I \in R/I$ such that $\phi_{sm}(r+I) = rsm = m$, i.e. $rs - 1 \in \text{Ann}(m) = I$, so $r+I$ and $s+I$ are inverse in R/I . Thus, R/I is a field, making I a maximal ideal.

Conversely, we show that R/I is a simple R -module if I is a maximal ideal of R . Since R/I is a field, multiplication by a nonzero element of R/I is a bijection on R/I , meaning that given $r+I \in R/I$, the set $R(r+I) = \{s(r+I) : s \in R\}$ is equal to R/I . In particular, if N is a nontrivial submodule of R/I containing a nonzero m , then $R/I = Rm \subseteq N$, so N is the entire module. $\square$

Problem 2. Suppose that M and N are submodules of a module P . Show that $(M+N)/M \cap N \cong (M/M \cap N) \oplus (N/M \cap N)$.

Solution. Let $\phi : (M/M \cap N) \oplus (N/M \cap N) \rightarrow (M+N)/M \cap N$ such that

$$\phi(a + M \cap N, b + M \cap N) = (a + b) + M \cap N.$$

This is of course surjective because any element of $M+N$ is written as $a+b$ for some $a \in M$, $b \in N$. To show it is injective, we need to show that if $a+b \equiv c+d \pmod{M \cap N}$ for $a, c \in M$ and $b, d \in N$, then $a \equiv c \pmod{M \cap N}$ and $b \equiv d \pmod{M \cap N}$. Indeed, we know a priori that $(a-c) + (b-d) \in M \cap N$, so since $a-c \in M$ and $(a-c) + (b-d) \in M$, their difference $b-d$ is also in M , hence in $M \cap N$, and similarly for $a-c$. Thus ϕ is an isomorphism between the two modules. $\square$

Problem 3. Suppose that R is an integral domain. An element $m \in M$ is said to be a **torsion element** if $rm = 0$ for some $r \neq 0$ in R , and we say that r **annihilates** m .

- (a) Let M_{tor} be the set of torsion elements of M . Show that M_{tor} is a submodule of M .
- (b) M is said to be **torsion-free** if $M_{\text{tor}} = 0$. Show that M/M_{tor} is always torsion-free.
- (c) Suppose $R = \mathbb{Z}$ and $M = \mathbb{R}/\mathbb{Z}$. What is M_{tor} in this case?

Solution. (a) Let $m_1, m_2 \in M_{\text{tor}}$, so there exist $r_1, r_2 \in R$ such that $r_1 m_1 = r_2 m_2 = 0$. Then for every $r \in R$, $r_1(r m_1) = r(r_1 m_1) = 0$, so M_{tor} is closed under multiplication by R , and $r_1 r_2(m_1 + m_2) = 0$, so M_{tor} is closed under addition.

(b) Let $m + M_{\text{tor}}$ be a torsion element of M/M_{tor} . Then $rm \in M_{\text{tor}}$ for some $r \in R$, so there exists $r' \in R$ such that $r'(rm) = (r'r)m = 0$, which in turn implies that $m \in M_{\text{tor}}$, i.e. $m + M_{\text{tor}}$ is the zero element of M/M_{tor} .

(c) The torsion submodule of $\mathbb{R}/\mathbb{Z}$ is $\mathbb{Q}/\mathbb{Z}$, since the torsion elements are the cosets corresponding to the elements of $\mathbb{R}$ which, when multiplied by some integer, are themselves integers. $\square$

Problem 4. Let M be a module and let $m \in M$. The **annihilator** $\text{Ann}(m)$ of m in R is the set $\{r \in R : rm = 0\}$.

- (a) Show that $\text{Ann}(m)$ is an ideal of R .
- (b) A module M is said to be **cyclic** if it is generated by a single element $m_0 \in M$. Show that M is cyclic if and only if $M \cong R/I$ for some ideal $I \subseteq R$.

Solution. (a) If $r_1, r_2 \in \text{Ann}(m)$ and $r \in R$ then

$$(r_1 + r_2)m = r_1m + r_2m = 0, \quad r(r_1m) = (rr_1)m = 0$$

implies that $\text{Ann}(m)$ is closed under addition and under multiplication by elements of R , hence is an ideal of R .

(b) Suppose M is generated by $m_0 \in M$ and let $I = \text{Ann}(m_0)$. Then, define $\phi: R/I \rightarrow M$ by $\phi(r + I) = rm_0$. This map is well-defined because if $r - s \in I$ then $rm_0 - sm_0 = (r - s)m_0 = 0$. It is surjective because M is generated by m_0 , and it is an injective map because

$$\phi(r + I) = rm_0 = 0 \implies r \in \text{Ann}(m) = I.$$

Thus ϕ is an isomorphism between R/I and M . □

Problem 5. Suppose that R is an integral domain, and let I be a nonzero ideal of R . Show that $I \cong R$ if and only if I is a principal ideal of R .

Solution. Suppose first that $I \cong R$ and let $\phi: R \rightarrow I$ be an isomorphism. Define $a = \phi(1)$. Then, any element of I can be written as $\phi(r) = r\phi(1) = ra$ for some $r \in R$, so $I = (a)$.

Conversely, if $I = (a)$ for some $a \in R$, then the morphism $\phi: R \rightarrow I$ determined by setting $\phi(1) = a$ is an isomorphism: it is clearly surjective because $I = \{ra = \phi(r) : r \in R\}$, and it is injective because $ra = r'a$ for $r, r' \in R$ implies $r = r'$, since R is an integral domain. □

Problem 6. Let F be a field and let V be an $F[x]$ -module that is finite-dimensional as a vector space over F .

- (a) Show that V is a torsion module.
- (b) Let $T: V \rightarrow V$ be the map defined by $T(v) = xv$ for $v \in V$. Show that T is a linear transformation of the F -vector space V .

Solution. (a) Since V is a finite-dimensional F -vector space, for each $v \in V$ there exists an n such that $v, xv, \dots, x^n v$ are F -linearly independent, so there exist $a_0, \dots, a_n \in F$, not all zero, such that

$$a_0v + a_1xv + \dots + a_nx^n v = (a_0 + a_1x + \dots + a_nx^n)v = 0,$$

which implies that v is a torsion element of V .

(b) T essentially inherits linearity from the fact that V is an $F[x]$ -module; for $v, w \in V$ and $c \in F$, $T(v + w) = x(v + w) = xv + xw = T(v) + T(w)$, and $T(cv) = x \cdot cv = c \cdot xv = cT(v)$. □

Problem 7. Let F be a field, G a finite group, and $F[G]$ the group ring. Let V be a finitely generated $F[G]$ -module.

- (a) Show that V is a finite-dimensional vector space over F .
- (b) Let $g \in G$, and let $T_g: V \rightarrow V$ be the map $T_g(v) = gv$. Show that T_g is a linear transformation of the F -vector space V .
- (c) Let $v_1, \dots, v_n$ be a basis for V over F , and for each $g \in G$, let M_g be the matrix of T_g with respect to the basis $v_1, \dots, v_n$. Show that $M_g \in GL_n(F)$ and that the map

$$\rho: G \rightarrow GL_n(F), \quad g \mapsto M_g$$

is a group homomorphism.

Solution. (a) Obviously V is a vector space over F ; the harder part of the problem is to show that this vector space is of finite dimension. Suppose V is generated by $v_1, \dots, v_m \in V$. Any element of V is of the form $r_1v_1 + \dots + r_mv_m$ where the r_i belong to $F[G]$, meaning $r_i = \sum_{g \in G} a_{i,g}g$. This means that any element of V is an F -linear combination of the elements of the finite set $\{gv_i : g \in G, i = 1, \dots, m\}$ (where the coefficients are the $a_{i,g}$), so V is spanned by a finite set, hence has finite dimension.

(b) As in the previous problem, linearity is pretty much trivial: For $v, w \in V$ and $c \in F$, $T_g(v + w) = g(v + w) = gv + gw = T_g(v) + T_g(w)$ and $T_g(cv) = g \cdot cv = c \cdot gv = cT_g(v)$.

(c) To show $M_g \in GL_n(F)$ we need to show that T_g is invertible; the inverse is simply $T_{g^{-1}}$, as $T_g(T_{g^{-1}}(v)) = gg^{-1}v = v$ and $T_{g^{-1}}(T_g(v)) = g^{-1}gv = v$. Then, to show that ρ is a homomorphism, we need to show that $T_{gh} = T_g \circ T_h$ for any $g, h \in G$; this is true because

$$T_{gh}(v) = ghv = T_g(hv) = T_g(T_h(v)) = (T_g \circ T_h)(v). \quad \square$$

Problem 8. Let R be a noetherian ring.

- (a) Let M be a submodule of R^n for some $n \geq 1$. Let $\pi_1: R^n \rightarrow R$ be the projection to the first coordinate, so $\pi_1(r_1, \dots, r_n) = r_1$. Show that $\pi_1(M)$ is an ideal in R .
- (b) Let $M_0 = \{m \in M : \pi_1(m) = 0\}$. Show that there is a finitely generated submodule $M' \subseteq M$ such that $M = M' + M_0$.
- (c) Prove that every submodule of R^n is finitely generated.
- (d) Deduce that every submodule of a finitely generated R -module is again finitely generated.

Solution. (a) $\pi_1(M)$ is a submodule of the R -module R , and submodules of rings are ideals.

(b) Since R is noetherian, $\pi_1(M)$ is finitely generated, say by $r_1, \dots, r_k \in \pi_1(M)$. Let $m_1, \dots, m_k \in M$ be such that $\pi_1(m_j) = r_j$ for $1 \leq j \leq k$, and let M' be the submodule generated by the m_j . Then, $\pi_1(M') = \pi_1(M)$ by construction, so if $m \in M$, there exists $m' \in M'$ such that the first coordinate of m is equal to that of m' . This means $\pi_1(m - m') = 0$, so $m - m' \in M_0$, proving that $M' + M_0 = M$.

(c) We induct on n , with the $n = 1$ case following from the fact that every submodule (ideal) of R is finitely generated because R is noetherian. In general, if $M \subseteq R^n$, then by (b), $M = M' + M_0$ where M' is finitely generated and $M_0 = \ker(\pi_1|_M)$ is isomorphic to a submodule of R , with embedding $M_0 \rightarrow R^{n-1}$ given by $(0, r_2, \dots, r_n) \mapsto (r_2, \dots, r_n)$. By inductive hypothesis, M_0 is finitely generated. To finish, any $m \in M$ can be written as $m' + m_0$ for some $m' \in M'$ and $m_0 \in M_0$, which

are respectively R -linear combinations of the generators of M' and M_0 , so the generators of M' and of M_0 together constitute a finite set of generators of M .

(d) Let M be a finitely generated R -module with generators $m_1, \dots, m_k$ and $N \subseteq M$ a submodule. Then, the morphism $\phi: R^k \rightarrow M$ given by $\phi(r_1, \dots, r_k) = r_1 m_1 + \dots + r_k m_k$ is surjective, so $M \cong R^k / \ker(\phi)$. Letting $\pi: R^k \rightarrow R^k / \ker(\phi)$ be the projection morphism, the submodule $\phi^{-1}(N)$ of $R^k / \ker(\phi)$ corresponding to N pulls back to a submodule $\pi^{-1}(\phi^{-1}(N)) \subseteq R^k$, which by (c) is finitely generated, say with generators $n_1, \dots, n_\ell$. Then, if $n \in \pi^{-1}(\phi^{-1}(N))$ is an R -linear combination of the n_j , its image $n + \ker(\phi)$ under π descends to an R -linear combination of the $n_j + \ker(\phi)$, so $\phi^{-1}(N)$ is generated by the finite set $\{n_j + \ker(\phi) : j = 1, \dots, \ell\}$, hence N is finitely generated. $\square$